AI Multi-Agent Travel Planner
Project Proposal

Abstract

The AI Multi-Agent Travel Planner is an intelligent travel planning system that uses multiple AI agents to collaboratively generate personalized travel itineraries. Instead of requiring users to search across multiple websites for flights, hotels, attractions, restaurants, weather, and budgets, the system coordinates specialized agents to provide a unified travel plan tailored to user preferences.

Problem Statement

Planning a trip is often time-consuming and fragmented. Users must compare flights, hotels, attractions, weather forecasts, and budgets across different platforms. Existing solutions typically focus on only one aspect of travel rather than delivering an end-to-end personalized experience. This project addresses that challenge through a collaborative multi-agent AI architecture.

Objectives

1. Build a multi-agent AI travel planner.
2. Generate personalized itineraries.
3. Recommend hotels, restaurants, and attractions.
4. Provide weather-aware suggestions.
5. Estimate and optimize travel budgets.
6. Present all recommendations through a simple conversational interface.

Stakeholders

Primary: Travelers, families, students, solo travelers, travel enthusiasts.

Secondary: Hotels, airlines, restaurants, tourism boards, travel agencies.

Inputs

Source, destination, travel dates, budget, number of travelers, interests, accommodation preference, dietary preference.

Outputs

Day-wise itinerary, hotel recommendations, attraction list, restaurant suggestions, weather forecast, estimated budget, packing checklist, travel tips, downloadable report.

Advantages

Personalized recommendations; saves time; centralized planning; scalable agent architecture; improved user experience; modular and extensible.

Limitations

Depends on internet connectivity and external APIs; pricing may change; limited by API quotas; recommendations depend on available data.

Scope

Current scope includes itinerary generation, hotel and attraction recommendations, weather integration, and budget estimation. Future scope includes booking integration, visa guidance, multilingual support, voice assistant, and AR navigation.

Technologies

Python, FastAPI, Streamlit, LangChain, LangGraph, Gemini/OpenAI APIs, ChromaDB/FAISS (optional), OpenWeather API, Google Maps API (optional).

System Architecture

User → Manager Agent → Flight, Hotel, Places, Food, Weather, Budget Agents → Aggregated Personalized Itinerary.

Expected Outcome

A practical GenAI application demonstrating AI agent orchestration, prompt engineering, and personalized decision-making for travel planning.

Conclusion

The project showcases modern Generative AI concepts using collaborating agents to automate complex travel planning while remaining modular and extensible.